



Fig. 2: Suggested enclosure

should then be opened).

When white LED is used as the bicycle headlight (with supporting optics), the result is quite good. However, the viewing angle and the brightness of LEDs vary inversely. A wider angle will result in lower brightness in any given direction as the total light output of the LED is spread over a larger area. For headlight applications, a 20°

viewing angle is appropriate. This narrow beam aspect of many of the white LEDs has a downside too. The narrower the LED beam, the dimmer it will appear when viewed off angle!

As shown in the circuit diagram, the headlight unit includes a 1W white LED and a small driver section. The driver section is, in fact, a constant-current driver realised using medium-power npn transistor BD139 (T1). The charger jack (J1) allows you to recharge the Ni-Cd battery pack with the help of an external DC power source such as a travel charger (9V) or a miniature solar panel (6V). The 12V zener diode (ZD1) in the circuit is added deliberately to protect the entire electronics (excluding the external charger) from an accidental over-voltage condition that may occur when the dynamo is running very fast.

Note. Bicycle dynamos are alternators equipped with permanent magnets. These are typically clawpole generators delivering energy at rather low RPM. The sidewall-running dynamo (*aka* bottle dynamo) is the most popular type.

Most bicycle dynamos are rated 6V, 3W (500 mA at 6V). However, the power of a dynamo light system depends somewhat on the speed. In a standard bicycle light system, the power is zero at standstill. At moderate speed it's about 3W, and at very high speed it's just a little more than 3W.

Some manufacturers include limiting zener diodes or other forms of overvoltage protection, to protect the light bulbs. Experienced hobbyists usually remove this internal protection circuitry to get more supply output from the same dynamo! ●

LED Voltmeter



RAJ K. GORKHALI

Four $\mu 741$ op-amps are used here to measure the voltage ranging from 3V to 12V DC. The voltage measured is displayed on eight LEDs.

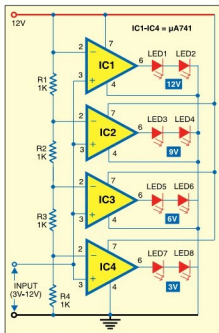
The inverting inputs (pin 2) of IC1 through IC4 are supplied with regulated 12V DC through voltage dividers R1 through R4. So the inverting input of each op-amp forms the reference volt-

age. The non-inverting inputs (pin 3) of the op-amps are tied together and connected to the input terminals. The voltage to be measured is fed to these input terminals.

When the input voltage exceeds the reference voltage of any of the op-amps, its output goes high, making the corresponding LEDs to glow.

This circuit is designed to measure 3V, 6V, 9V and 12V DC. For instance, when the input voltage is 3V, LED7 and LED8 glow. When the input voltage is 12V, all the LEDs glow.

Note. You can use a quad device such as LM324 instead of $\mu 741$ ICs. ●



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